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CM3070 FINAL REPORT

DEEP LEARNING BREAST CANCER DETECTION

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CHAPTER 1: INTRODUCTION (975/1000 words)

1.1 Project Overview

This project is based on Project Idea 2: Deep Learning Breast Cancer Detection, one of the suggested project templates for the CM3070 Final Project module. It develops and evaluates deep learning models for classifying full mammograms from the CBIS-DDSM dataset as benign or malignant.

The project follows an iterative experimental workflow involving dataset preparation, exploratory data analysis, patient-level data splitting, image preprocessing, custom CNN development, transfer learning, fine-tuning, and performance evaluation. Three custom CNN configurations with different numbers of convolutional blocks are investigated alongside ResNet50, EfficientNetB0, and DenseNet121. All models use consistent dataset partitions and evaluation procedures to support controlled comparison.

The evaluation extends beyond overall accuracy by considering precision, recall or sensitivity, specificity, F1-score, AUC, confusion matrices, and false-positive and false-negative predictions. The strongest fine-tuned model is further investigated through classification-threshold optimisation using validation data. Grad-CAM is applied to representative predictions to examine the image regions influencing model behaviour and identify possible reliance on non-clinical features. Together, these experiments investigate how architecture, fine-tuning, and threshold selection affect mammography classification performance within the scope of CBIS-DDSM.

1.2 Background

Breast cancer is one of the most common cancers affecting women worldwide and remains a major cause of cancer-related deaths. According to the World Health Organization (2024), early detection and diagnosis can significantly improve treatment outcomes and survival rates. Mammography is widely used for breast cancer screening because it enables abnormalities and potential signs of breast cancer to be identified before symptoms become apparent.

Despite its importance, mammogram interpretation is challenging. Radiologists are required to examine large numbers of medical images, while diagnostic accuracy may be influenced by factors such as image quality, workload, and human fatigue. Classification errors can also have important consequences. False-positive findings may lead to unnecessary further examinations, biopsies, and patient anxiety, whereas false-negative findings may delay diagnosis and subsequent treatment.

Recent advances in artificial intelligence (AI), particularly deep learning, have demonstrated significant potential in medical image analysis. CNNs can automatically learn hierarchical visual features from images rather than relying on manually designed features. This capability has led to their increasing application in medical image classification and computer-aided diagnosis (CAD), where AI systems may provide additional decision support to healthcare professionals.

This project investigates custom CNNs alongside three transfer-learning architectures: ResNet50 (He et al., 2016), EfficientNetB0 (Tan and Le, 2019), and DenseNet121 (Huang et al., 2017). The custom CNNs provide task-specific baselines with different depths, while the pretrained architectures introduce residual connections, compound scaling, and dense connectivity. Evaluating them under a common methodology enables their relative effectiveness and error patterns to be assessed on the same classification task.

1.3 Problem Statement

Breast cancer screening relies on mammography and expert interpretation by radiologists. However, manual interpretation is demanding, and diagnostic performance can vary with image complexity and reader experience. Artificial intelligence therefore has potential as a computer-aided approach for supporting mammography analysis (Rodriguez-Ruiz et al., 2019).

Although deep learning has achieved promising results in medical image analysis, performance is influenced by architecture, dataset characteristics, preprocessing, training strategy, and evaluation design (Litjens et al., 2017). Results from different studies are difficult to compare directly when they use different datasets, partitions, preprocessing procedures, and metrics. Furthermore, overall accuracy may conceal clinically important differences in malignant-case sensitivity and false-negative predictions.

A controlled comparison is therefore needed to examine not only how architectures perform under consistent conditions, but also how model refinement and decision thresholds affect their error trade-offs. This project addresses this problem by evaluating custom CNN configurations and pretrained architectures using the same patient-level partitions and evaluation framework. The transfer-learning models are subsequently fine-tuned, and the strongest model is evaluated using a threshold selected exclusively from validation data. This enables the project to investigate whether improved malignant-case detection can be achieved without treating accuracy as the sole criterion.

1.4 Aim

The aim of this project is to develop, refine, and evaluate deep learning models for classifying full mammograms as benign or malignant, with particular attention to malignant-case sensitivity and false-negative reduction.

1.5 Objectives

To achieve this aim, the project has the following objectives:

1. Investigate deep learning techniques and methodological considerations relevant to automated mammography classification.
2. Prepare the CBIS-DDSM dataset using consistent preprocessing and patient-level training, validation, and testing partitions to reduce data-leakage risk.
3. Develop and compare custom CNN configurations and transfer-learning implementations of ResNet50, EfficientNetB0, and DenseNet121.
4. Fine-tune the three transfer-learning models and evaluate all models using accuracy, precision, recall or sensitivity, specificity, F1-score, AUC, ROC curves, confusion matrices, and classification-error analysis.
5. Optimise the classification threshold of the strongest fine-tuned model using validation data and examine its effect on malignant recall, false negatives, false positives, and specificity.
6. Use representative error cases and Grad-CAM to investigate model behaviour, possible spurious attention, and limitations relevant to computer-aided mammography classification.

These objectives progress from establishing a reliable dataset and experimental pipeline to refining and critically evaluating model performance. Fine-tuning, validation-based threshold selection, error analysis, and explainability provide a focused contribution beyond a conventional comparison of established architectures.

1.6 Research Question

How do different deep learning approaches and validation-based threshold optimisation affect overall classification performance and malignant-case detection in CBIS-DDSM full mammograms?

1.7 Deliverables

The project deliverables are:

- A critical literature review of deep learning approaches and methodological considerations for mammography classification.

- A prepared CBIS-DDSM dataset with patient-level training, validation, and testing partitions.
- Three custom CNN configurations for investigating the effect of network depth.
- Frozen and fine-tuned implementations of ResNet50, EfficientNetB0, and DenseNet121.
- A comparative evaluation using accuracy, precision, recall or sensitivity, specificity, F1-score, AUC, ROC curves, confusion matrices, and false-positive and false-negative analysis.
- Validation-based threshold analysis for the strongest fine-tuned model.
- Grad-CAM visualisations and analysis of representative correct and incorrect predictions.
- A critical discussion of performance, limitations, and potential computer-aided applications.
- A final report and 3 to 5 minute demonstration video documenting the implementation, evaluation, findings, and project outcomes.

CHAPTER 2: LITERATURE REVIEW (2287/2500 words)

2.1 Deep Learning in Medical Image Analysis

Deep learning has become an important approach in medical image analysis because it can automatically learn complex visual features directly from images. Unlike traditional machine learning methods that often rely on manually designed features, deep learning can learn hierarchical representations suitable for classification, detection, and segmentation.

Litjens et al. (2017) reviewed more than 300 medical imaging studies and demonstrated the widespread application of Convolutional Neural Networks (CNNs). However, they also identified limitations involving labelled data availability and the ability of models to generalise across datasets and clinical settings. Similarly, Shen et al. (2017) highlighted the potential of deep learning for computer-aided diagnosis (CAD), while identifying challenges involving data availability, computational requirements, and model interpretability.

More recently, Carriero et al. (2024) reviewed deep learning in breast imaging and reported continued developments across mammography, ultrasound, and MRI. Nevertheless, challenges involving explainability, reproducibility, external validation, and clinical integration remain. Consequently, strong performance in an individual study does not necessarily indicate reliable performance under different datasets or experimental conditions.

Overall, the literature demonstrates the potential of deep learning for medical and breast image analysis while also showing that performance is influenced by architectural and methodological choices. This supports the need to evaluate different deep learning approaches under consistent experimental conditions rather than assuming that a single architecture is universally superior.

2.2 Deep Learning Architectures for Mammography Classification

2.2.1 Baseline CNN

CNNs are widely used for image classification because convolutional filters can learn spatial features such as edges, textures, and increasingly complex image representations. This is relevant to mammography, where benign and malignant tissue may differ through subtle visual patterns.

Nasser and Yusof (2023) reviewed deep learning approaches for breast cancer diagnosis and identified CNN-based architectures as widely applied to breast imaging. However, reported performance varied according to network architecture, dataset characteristics, preprocessing, and experimental methodology.

A custom CNN therefore provides a useful baseline against which transfer learning can be assessed. It learns features directly from the project dataset, but this may also limit performance because the available mammography data are considerably smaller than datasets commonly used to pretrain transfer learning models. The baseline CNN consequently provides a reference for determining whether more advanced architectures offer meaningful improvements.

2.2.2 ResNet50

ResNet50 was introduced by He et al. (2016) to address difficulties associated with training increasingly deep neural networks. Residual connections allow information and gradients to flow more effectively across layers, enabling deeper networks to be optimised successfully.

Greater depth may enable complex hierarchical features to be learned from mammograms. Nasser and Yusof (2023) identified ResNet-based approaches among architectures applied to breast cancer diagnosis. However, greater depth does not guarantee improved classification and also increases computational requirements compared with a small custom CNN. Performance remains influenced by the dataset, preprocessing, and transfer learning strategy. ResNet50 is therefore included to evaluate whether residual learning and deeper feature extraction improve performance over the baseline CNN.

2.2.3 EfficientNetB0

Tan and Le (2019) introduced EfficientNetB0 to balance classification performance and computational efficiency. Its compound scaling method systematically scales network depth, width, and input resolution together rather than modifying one dimension independently.

This approach is relevant to medical imaging, where processing large image datasets can require substantial computational resources. Nasser and Yusof (2023) identified EfficientNetB0 based approaches among architectures applied to breast cancer diagnosis. However, computational efficiency does not guarantee superior mammography classification, as performance remains dependent on image resolution, preprocessing, dataset characteristics, and training strategy. EfficientNetB0 is therefore included to investigate whether an efficiency-oriented architecture can achieve competitive classification performance.

2.2.4 DenseNet121

DenseNet121 was introduced by Huang et al. (2017) and uses dense connectivity to improve information flow between layers. Within dense blocks, previously learned feature maps are reused by subsequent layers, encouraging feature reuse and improving gradient propagation.

This may benefit medical image classification because both lower- and higher-level features can contribute to identifying abnormalities. However, DenseNet121 does not necessarily provide superior performance, and maintaining connections between numerous layers can increase memory requirements during training. DenseNet121 is therefore included to investigate whether feature reuse provides advantages for mammography classification compared with the other selected architectures.

2.2.5 Comparative Analysis of Selected Architectures

The selected architectures represent different approaches to feature learning and model complexity. Table 2.1 summarises their principal differences.

Table 2.1. Comparison of Selected Deep Learning Architectures.

Architecture	Main Feature	Potential Advantage	Potential Limitation	Role
Baseline CNN	Conventional convolutional blocks	Simple architecture that learns features directly from the project dataset	Limited depth and no pretrained feature representations	Provides a benchmark for comparison
ResNet50	Residual connections	Supports deeper feature extraction and improves information and gradient flow	Greater computational complexity than a small custom CNN	Evaluates the benefits of residual learning and greater network depth
EfficientNetB0	Compound scaling of depth, width and resolution	Designed to balance predictive performance and computational efficiency	Performance remains dependent on dataset and training strategy	Evaluates whether an efficiency-oriented architecture provides competitive performance
DenseNet121	Dense connectivity and feature reuse	Encourages feature reuse and strong	Dense connectivity can increase memory	Evaluates whether feature reuse benefits

		information flow between layers	requirements during training	mammography classification

Table 2.1 shows that the architectures offer different potential advantages. The custom CNN provides a task-specific benchmark but lacks pretrained features. ResNet50 supports deeper feature extraction through residual connections, EfficientNetB0 balances capacity and computational efficiency through compound scaling, and DenseNet121 encourages feature reuse through dense connectivity.

However, architectural complexity does not guarantee superior mammography classification. Performance also depends on the dataset, preprocessing, class distribution, and training strategy. Evaluating the architectures using the same CBIS-DDSM partitions and methodology therefore provides a controlled assessment of whether residual learning, compound scaling, or dense connectivity offers an advantage under the project's experimental conditions.

2.3 Deep Learning Studies in Breast Cancer Detection

Deep learning has been investigated for mammography applications including detection, segmentation, classification, and risk assessment. Wang (2024) reviewed deep learning-assisted mammography and identified improvements across these applications, while also highlighting limitations involving interpretability, dataset diversity, privacy, and generalisation.

Similarly, Carriero et al. (2024) reported widespread use of CNN-based approaches in breast imaging but identified continuing challenges involving explainability, reproducibility, external validation, and clinical integration. These findings demonstrate that predictive performance must be considered alongside the reliability and generalisability of experimental methodology.

Unlike these reviews, Ahmad et al. (2024) developed a multi-stage CAD framework using CBIS-DDSM and reported 98.5% detection accuracy, 95.39% segmentation performance, and 99.16% classification accuracy. However, these results should not be directly compared with simpler classification models because the task, architecture, preprocessing, and experimental methodology differ. The authors also identified high computational requirements associated with the large number of trainable parameters.

Table 2.2. Comparison of Selected Deep Learning Studies in Breast Cancer Detection.

Study	Dataset	Approach	Reported Performance	Key Limitations
Wang	Multiple datasets	CNN, ResNet50,	N/A	Differences in

(2024)		EfficientNetB0, DenseNet and other approaches		datasets and evaluation methodologies limit direct comparison
Carriero et al. (2024)	Multiple datasets	Review of deep learning approaches for breast imaging	N/A	Explainability, reproducibility, external validation, and clinical integration remain challenges
Ahmad et al. (2024)	CBIS-DDSM	Multi-stage CAD framework for detection, segmentation, and classification	Classification: 99.16%; Detection: 98.5%; Segmentation: 95.39%	High model complexity and computational requirements

Table 2.2 demonstrates that reported performance cannot be compared through accuracy alone because studies differ in datasets, tasks, preprocessing, and evaluation procedures. For example, Ahmad et al. (2024) combined detection, segmentation, and classification, making its results unsuitable for direct comparison with a whole-image binary classifier. This limitation motivates the present controlled comparison, in which all architectures use the same CBIS-DDSM preparation, partitions, and evaluation framework.

2.4 Data Preparation Challenges in Mammography

2.4.1 Image Preprocessing and Data Augmentation

Preprocessing can directly affect features learned from mammography images. Common approaches include resizing, normalisation, contrast enhancement, noise reduction, and cropping (Abdelhafiz et al., 2019). Resizing high-resolution mammograms reduces computational requirements but may also remove fine information associated with subtle abnormalities, creating a trade-off between image detail and computational feasibility.

Data augmentation can increase training diversity and reduce overfitting when labelled data are limited (Shorten and Khoshgoftaar, 2019). Mammography studies commonly use transformations including rotation, flipping, and scaling (Abdelhafiz et al., 2019). However, augmentation should preserve clinically relevant image characteristics; inappropriate transformations may introduce unrealistic variation.

A consistent preprocessing and augmentation pipeline is therefore important when comparing architectures, as it reduces the likelihood that performance differences arise from different input preparation rather than the models themselves.

2.4.2 Class Imbalance

Class imbalance can cause classification models to favour the majority class, meaning overall accuracy may conceal poor performance on less represented cases. This is particularly important in breast cancer classification because failure to identify malignant cases is a clinically significant error. Metrics such as precision, recall, F1-score, and AUC therefore provide additional information beyond accuracy (Aggarwal et al., 2021).

Imbalance can be addressed using techniques including undersampling, oversampling, augmentation, and class weighting, although each involves trade-offs. Jaamour et al. (2023) investigated class weighting using CBIS-DDSM and demonstrated that increasing minority-class weighting does not automatically improve classification performance. This illustrates that imbalance-handling strategies must be evaluated rather than assumed to be beneficial.

Consequently, this project considers multiple performance metrics and confusion matrices when comparing models, particularly to assess their ability to identify malignant cases.

2.4.3 Patient-Level Data Splitting

Data splitting is particularly important in mammography because multiple images, including craniocaudal (CC) and mediolateral oblique (MLO) views, may originate from the same patient. Randomly splitting individual images can therefore place images belonging to one patient in both training and testing sets.

This creates a risk of data leakage because images from the same patient may share anatomical and imaging characteristics, potentially producing an overly optimistic estimate of generalisation. Patient-level splitting addresses this by assigning all images associated with one patient exclusively to a single training, validation, or testing partition.

Patient-level separation may make it more difficult to maintain equivalent class distributions across partitions, so stratification should be considered where possible. Nevertheless, it provides a more reliable assessment of performance on previously unseen patients and ensures that all models are evaluated using the same independent cases.

2.5 CBIS-DDSM Dataset

The Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM) is a publicly available mammography dataset developed to provide a more standardised version of the original Digital Database for Screening Mammography (DDSM). Lee et al. (2016) describe CBIS-DDSM as containing selected and curated mammography cases with updated image formats and improved region-of-interest

segmentation and bounding boxes. The dataset includes mammography images together with associated lesion and pathology information, making it suitable for supervised breast imaging research.

A major advantage of CBIS-DDSM is that it provides labelled cases that can support the development and evaluation of deep learning models for tasks including breast abnormality classification. Its public availability also improves reproducibility and enables researchers to evaluate different approaches using a common dataset. This makes CBIS-DDSM particularly useful for comparative studies where multiple architectures need to be assessed under consistent experimental conditions.

However, using CBIS-DDSM also introduces methodological challenges. Mammography images are high resolution, creating substantial computational requirements and often requiring resizing or other preprocessing before they can be used with deep learning architectures. The dataset also contains multiple images and mammographic views associated with individual patients, making patient-level separation important when constructing training, validation, and testing sets. Furthermore, differences in class distribution need to be considered because they may influence model training and the interpretation of overall classification accuracy.

Despite these limitations, CBIS-DDSM provides an appropriate dataset for the objectives of this project because it combines publicly available mammography images with pathology information required for supervised classification. More importantly, using a single dataset and consistent data partitions across the baseline CNN, ResNet50, EfficientNetB0, and DenseNet121 provides a controlled basis for comparing the selected architectures. The methodological challenges associated with the dataset are therefore addressed through the preprocessing, class-imbalance considerations, and patient-level data separation discussed in Section 2.4.

2.6 Research Gap and Project Motivation

The literature demonstrates that deep learning has considerable potential for automated breast cancer classification, with CNN-based and transfer learning approaches achieving promising results across mammography studies. However, the review also shows that no single architecture can be considered universally superior. ResNet50, EfficientNetB0, and DenseNet121 employ different strategies for feature extraction and information flow, while their effectiveness remains dependent on the dataset and experimental methodology used.

A significant limitation when comparing existing studies is the variation in datasets, preprocessing procedures, model configurations, data-splitting strategies, and evaluation metrics. Consequently, differences in reported performance cannot always be attributed

solely to model architecture. In addition, methodological factors such as class imbalance and patient-level data separation can influence the reliability of evaluation results. These issues make controlled comparisons particularly important when assessing deep learning approaches for mammography classification.

Nasser and Yusof (2023) similarly identified the need for comprehensive evaluation of deep learning approaches for breast cancer diagnosis. Although previous studies demonstrate the potential of advanced architectures, further comparison under consistent experimental conditions can provide clearer evidence of their relative strengths and limitations. Evaluation should also extend beyond overall accuracy, particularly where unequal class distributions may conceal differences in the detection of malignant cases.

To address these issues, this project evaluates three custom CNN configurations alongside frozen and fine-tuned versions of ResNet50, EfficientNetB0, and DenseNet121 using consistent CBIS-DDSM patient-level partitions and evaluation procedures. The comparison is extended through validation-based threshold selection for the strongest fine-tuned model, with particular attention to malignant-case sensitivity and false-negative reduction. Representative error analysis and Grad-CAM are then used to investigate prediction confidence and possible reliance on non-clinical image features. This provides a controlled assessment of how architecture, fine-tuning, and operating-threshold selection affect mammography-classification performance.

CHAPTER 3:PROJECT DESIGN (1562/2000 words)

3.1 Domain and Intended Users

This project is situated within medical-image analysis and computer-aided diagnosis. It investigates deep learning for classifying CBIS-DDSM full mammograms as benign or malignant.

Potential users would include radiologists and breast-imaging specialists. A system based on this research could provide a malignant-case probability as additional decision support rather than replace professional judgement. However, the developed system is an experimental prototype and has not been clinically validated.

Custom CNNs and pretrained architectures are evaluated using multiple metrics, with particular attention to sensitivity and false negatives. Threshold analysis, error analysis, and Grad-CAM are used to investigate the final model's performance and limitations.

3.2 Design Rationale

The project was designed as a controlled comparative study in which multiple deep learning approaches are trained and evaluated using common dataset partitions and evaluation procedures. This design was selected because results reported across independent mammography studies are difficult to compare when different datasets, preprocessing methods, data-splitting strategies, and metrics are used.

Custom CNNs provide task-specific baselines that learn features directly from the CBIS-DDSM training data. Rather than relying on one baseline configuration, CNNs containing two, three, and four convolutional blocks are examined to investigate whether increased network depth improves classification performance. Batch normalisation, dropout, data augmentation, class weighting, and training callbacks are used to improve training stability and reduce overfitting.

ResNet50, EfficientNetB0, and DenseNet121 were selected because they represent different approaches to deep feature extraction. ResNet50 uses residual connections, EfficientNetB0 applies compound scaling, and DenseNet121 promotes feature reuse through dense connectivity. Each architecture is initially used as a frozen feature extractor. Selected upper layers are then unfrozen and trained at a lower learning rate, while Batch Normalisation layers remain frozen to preserve their pretrained statistics.

All models use the same patient-level training, validation, and testing partitions. Architecture-specific preprocessing is applied where required, while augmentation is restricted to the training partition. Class weighting is used to reduce the effect of the moderate imbalance between benign and malignant images.

The design further recognises that a probability-based classifier can produce different error patterns depending on its decision threshold. The strongest fine-tuned model is therefore subjected to validation-based threshold analysis. This extends the research beyond architecture comparison by investigating whether malignant-case sensitivity and false-negative performance can be improved through a justified operating threshold.

3.3 System Architecture

The project follows a supervised deep learning workflow for classifying full mammography from CBIS-DDSM. Figure 3.1 presents the system architecture, which consists of the following stages:

- Acquisition and preparation of CBIS-DDSM metadata and mammography images.
- Exploratory Data Analysis to examine class, lesion-type, mammographic-view, and patient distributions.
- Extraction of pathology labels, image paths, and patient identifiers.
- Stratified patient-level division into training, validation, and testing partitions.
- Image resizing, normalisation, and architecture-specific preprocessing.
- Data augmentation and class weighting during training.
- Iterative development of two-, three-, and four-block custom CNNs.
- Frozen-backbone transfer learning using ResNet50, EfficientNetB0, and DenseNet121.
- Fine-tuning of selected upper layers in all three pretrained networks.
- Comparative evaluation using classification metrics, ROC curves, and confusion matrices.
- Validation-based threshold selection for the strongest fine-tuned model.
- Analysis of false-positive and false-negative cases.
- Grad-CAM analysis of representative correct and incorrect predictions.

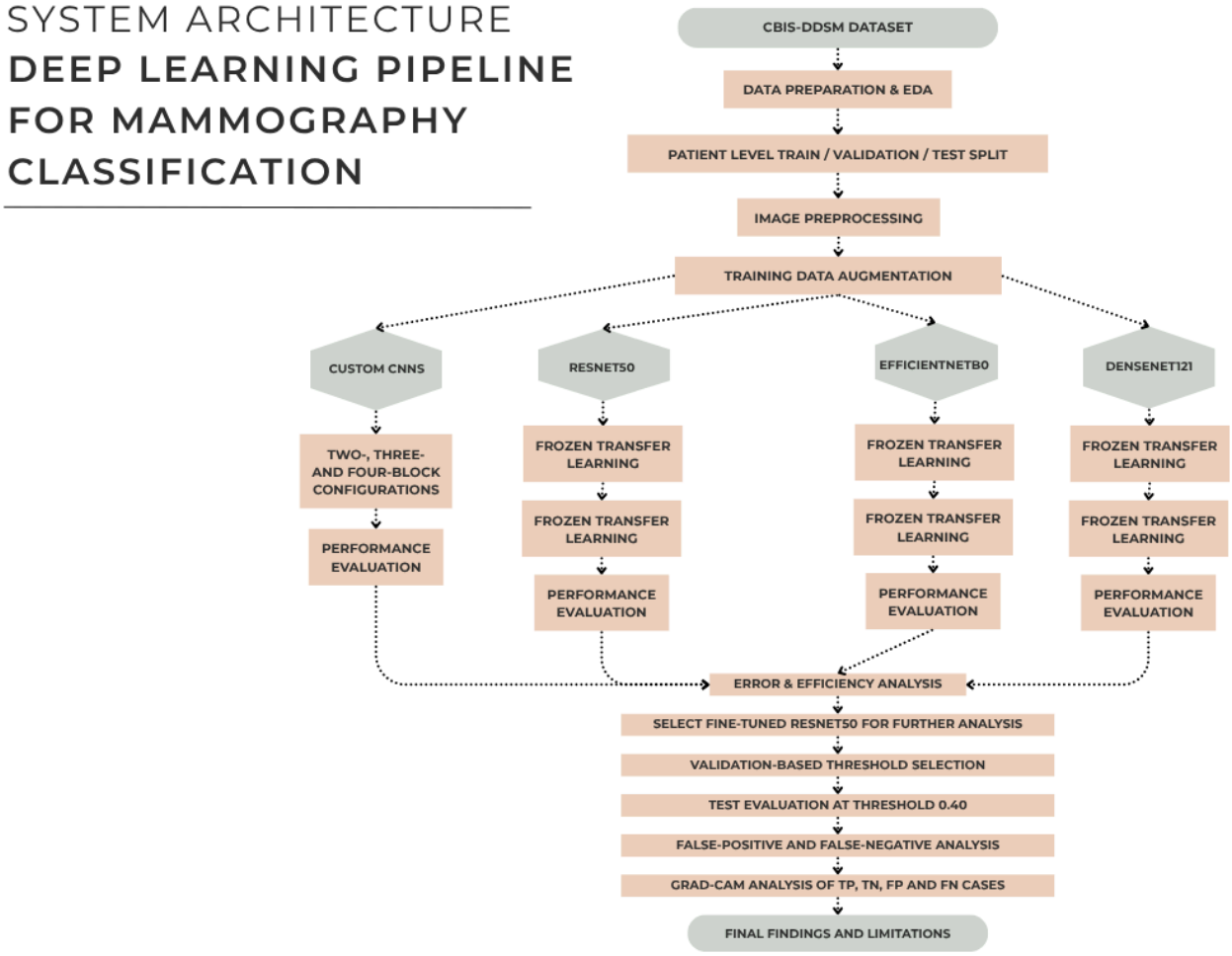


Figure 3.1. Deep learning pipeline for mammography classification.

Patient-level separation ensures that images belonging to the same patient do not appear in multiple partitions, reducing the risk of data leakage. Data augmentation is applied only to training images so that validation and test performance is measured using unchanged samples.

After the initial comparison, fine-tuning is used to assess whether the pretrained features can be adapted more effectively to mammography images. The strongest resulting model is then investigated through threshold selection, error analysis, and Grad-CAM. The architecture therefore progresses from controlled model comparison to a more detailed examination of malignant-case detection and model behaviour.

3.4 Technologies and Model Selection

3.4.1 Dataset

The project uses the Curated Breast Imaging Subset of the Digital Database for Screening Mammography. CBIS-DDSM provides mammography images with pathology labels and lesion information suitable for supervised classification. Its public availability also supports reproducibility and comparison with related research.

3.4.2 Development Environment

Python is used as the primary programming language because of its support for machine learning, image processing, data analysis, and scientific computing. Development and experimentation are conducted in Google Colab, which provides a notebook-based environment and access to GPU acceleration. Google Drive is used to store the dataset, trained models, and experiment outputs.

3.4.3 Software Libraries

TensorFlow and Keras are used to construct, train, fine-tune, and evaluate the deep learning models (TensorFlow, n.d.; Keras Team, n.d.). Pandas and NumPy support metadata processing, dataset preparation, and numerical operations. Scikit-learn is used to calculate class weights and evaluation metrics, while Matplotlib is used to visualise dataset distributions, training histories, confusion matrices, ROC curves, model comparisons, and Grad-CAM results.

3.4.4 Model Selection

The experimental design includes:

- Two-, three-, and four-block regularised custom CNNs.
- ResNet50.
- EfficientNetB0.
- DenseNet121.

The custom CNN configurations investigate the effect of convolutional depth and establish task-specific baselines. ResNet50 investigates residual learning, EfficientNetB0 examines an efficiency-oriented scaling strategy, and DenseNet121 examines dense connectivity and feature reuse. The three pretrained models are evaluated before and after fine-tuning, enabling the effect of domain adaptation to be assessed separately from architectural differences.

3.5 Implementation Methodology

Phase 1: Dataset preparation and partitioning

Pathology labels are converted into binary classes, where benign and benign-without-callback cases are labelled as benign and malignant cases are labelled as malignant. Full mammogram records are matched with their metadata and checked for valid file paths. Patients are then divided into approximately 70% training, 15% validation, and 15% testing partitions using stratification. All images belonging to one patient remain within the same partition.

Phase 2: Preprocessing and imbalance handling

Images are resized to 224 × 224 pixels and converted to three-channel input. Pixel rescaling is used for the custom CNNs, while the transfer-learning models receive their corresponding architecture-specific preprocessing. Rotation of up to 10 degrees, horizontal flipping, and zooming of up to 10% are applied only to the training data. Class weights calculated from the training labels assign greater importance to malignant cases.

Phase 3: Iterative CNN Development

Custom CNNs containing two, three, and four convolutional blocks are trained under consistent conditions. Batch normalisation, max pooling, dropout, and training callbacks are incorporated. This experiment examines whether increasing convolutional depth produces better generalisation.

Phase 4: Transfer Learning & Fine-Tuning

ResNet50, EfficientNetB0, and DenseNet121 are initialised using ImageNet weights with their original classification heads removed. Their backbones are initially frozen, and a common head containing Global Average Pooling, dropout, and sigmoid classification is trained. Selected upper layers are subsequently unfrozen and fine-tuned using a learning rate of $1\ x\ 10^{-5}$. Batch Normalisation layers remain frozen during this stage.

Phase 5: Model Comparison

Frozen and fine-tuned models are evaluated using the same test partition and compared using multiple performance measures. This determines how fine-tuning affects each architecture and identifies the model showing the strongest overall combination of discrimination and malignant-case detection.

Phase 6: Threshold Selection

Candidate thresholds from 0.30 to 0.70 are assessed using the validation partition.

Sensitivity, specificity, precision, F1-score, and Youden's J statistic are calculated for each threshold. A threshold of 0.40 is selected because it produces the highest Youden's J statistic, representing the best observed balance between sensitivity and specificity. This fixed threshold is then applied to the fine-tuned ResNet50 test probabilities and compared with the default threshold of 0.50.

Phase 7: Error Analysis & Explainability

False-positive and false-negative predictions from the final ResNet50 configuration are identified and inspected. Grad-CAM is generated for representative true-positive, true-negative, false-positive, and false-negative predictions. The heatmaps are used to investigate whether the model focuses on plausible breast regions or potentially spurious features such as image borders and embedded markers.

3.6 Model Evaluation Design

The malignant class is treated as the positive class. The following measures are used:

$$Accuracy = \frac{No.\ of\ Correct\ Predictions}{Total\ No.\ of\ Predictions}$$

Accuracy measures the proportion of all images classified correctly but may conceal differences between benign- and malignant-case performance.

$$Precision = \frac{TP}{TP + FP}$$

(Where TP = True Positives, FP = False Positives)

Precision measures the proportion of predicted malignant images that are genuinely malignant.

$$Sensitivity = \frac{TP}{TP + FN}$$

(Where FN = False Negatives)

Sensitivity, also referred to as malignant recall, measures the proportion of malignant images correctly identified. It is particularly important because false negatives represent malignant cases classified as benign.

$$Specificity = \frac{TN}{TN + FP}$$

Specificity measures the proportion of benign images correctly classified.

$$F1\ Score\ =\ 2\ X\ \frac{Precision\ X\ Recall}{Precision + Recall}$$

The F1-score combines malignant precision and sensitivity and is useful when both false positives and false negatives must be considered.

AUC measures the model's ability to rank malignant images above benign images across possible thresholds. Unlike threshold-dependent metrics, it evaluates the predicted probabilities rather than classifications produced at one operating threshold.

Confusion matrices report true positives, true negatives, false positives, and false negatives. ROC curves show the relationship between sensitivity and false-positive rate across thresholds. Model-parameter counts and observed training behaviour are also considered when interpreting differences between architectures.

3.7 Project Schedule

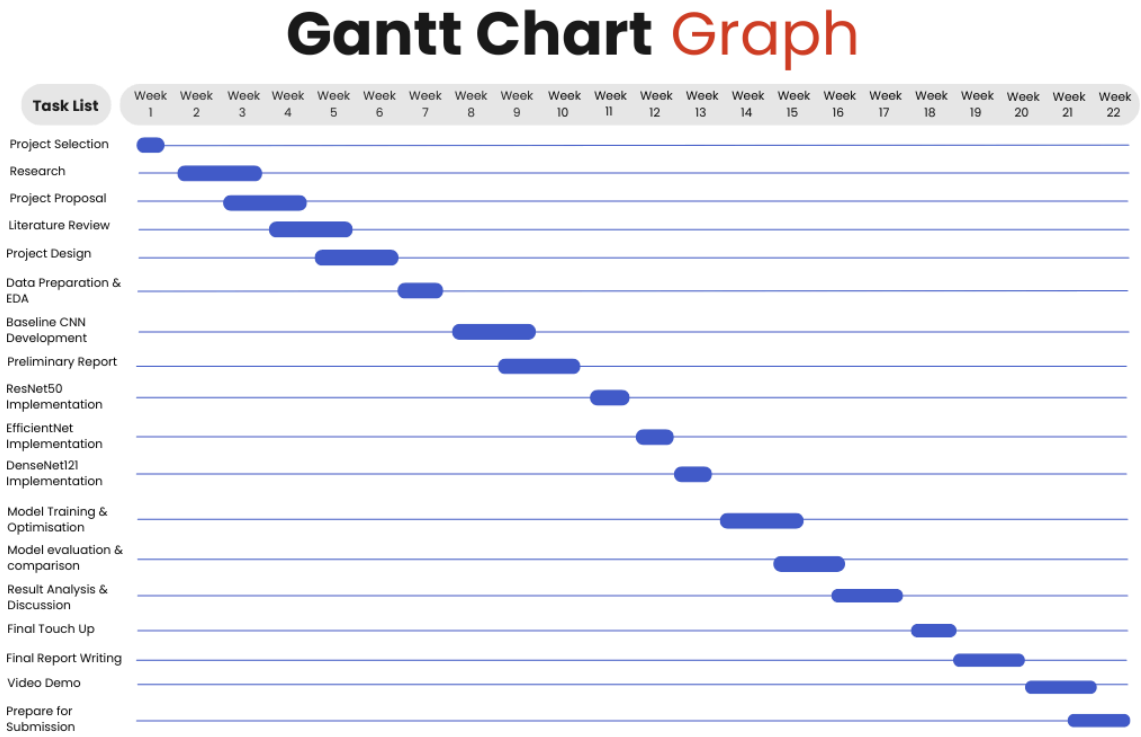


Figure 3.2. Project schedule and major development milestones.

The schedule progressed from research and project design to dataset preparation, model development, fine-tuning, evaluation, and report completion. Its iterative structure allowed results to inform later threshold analysis, error investigation, and Grad-CAM before preparation of the final report and demonstration video.

CHAPTER 4: IMPLEMENTATION (2452/2500 words)

4.1 Implementation Overview

The project was implemented in Google Colab using Python, TensorFlow, and Keras. The pipeline covered dataset preparation, exploratory analysis, patient-level splitting, preprocessing, model development, fine-tuning, and evaluation.

Two-, three-, and four-block custom CNNs were compared with ResNet50, EfficientNetB0, and DenseNet121. The pretrained backbones were initially frozen before selected upper layers were fine-tuned. Fine-tuned ResNet50 was subsequently selected for validation-based threshold analysis, classification-error analysis, and Grad-CAM.

4.2 Dataset Preparation and Exploratory Data Analysis (EDA)

The CBIS-DDSM dataset was stored in Google Drive and accessed through Google Colab. The implementation loaded the mass and calcification case-description files together with dicom_info.csv, which described the available image files and image types.

Initial exploration of the raw case-description metadata identified 3,568 abnormality-level records. These comprised 2,111 benign records, including cases labelled benign without callback, and 1,457 malignant records. This distribution showed that benign cases formed the majority class before image matching, although the imbalance was not extreme.

The image metadata contained 3,567 cropped-image records, 3,247 ROI-mask records, and 2,857 full-mammogram records. Cropped images and ROI masks were excluded because the project investigated whole-image classification. Retaining only full mammograms also prevented the model from receiving explicitly localised lesion regions that would not normally be available before diagnosis.

Pathology labels were converted into binary values. Malignant cases were assigned a label of 1, while benign and benign-without-callback cases were assigned a label of 0. The case-description metadata were then matched with the corresponding full-mammogram paths using image identifiers. All matched paths were checked to confirm that their image files existed.

Following matching, the modelling dataframe contained 3,286 lesion-linked records from 1,460 patients. Of these records, 1,931 were labelled benign and 1,355 were labelled malignant, representing approximately 58.76% and 41.24% of the matched dataset respectively. Because the source case-description files contained separate records for annotated abnormalities, one full mammogram associated with multiple abnormalities could contribute more than one matched record.

Representative mammograms were examined to verify image loading and explore visual differences across benign and malignant mass and calcification cases. The images varied in breast size, orientation, tissue appearance, contrast, and the amount of surrounding black background. Text labels and acquisition markers were also visible in some images. These observations were relevant because inconsistent positioning and non-clinical markers could potentially influence the features learned by the models.

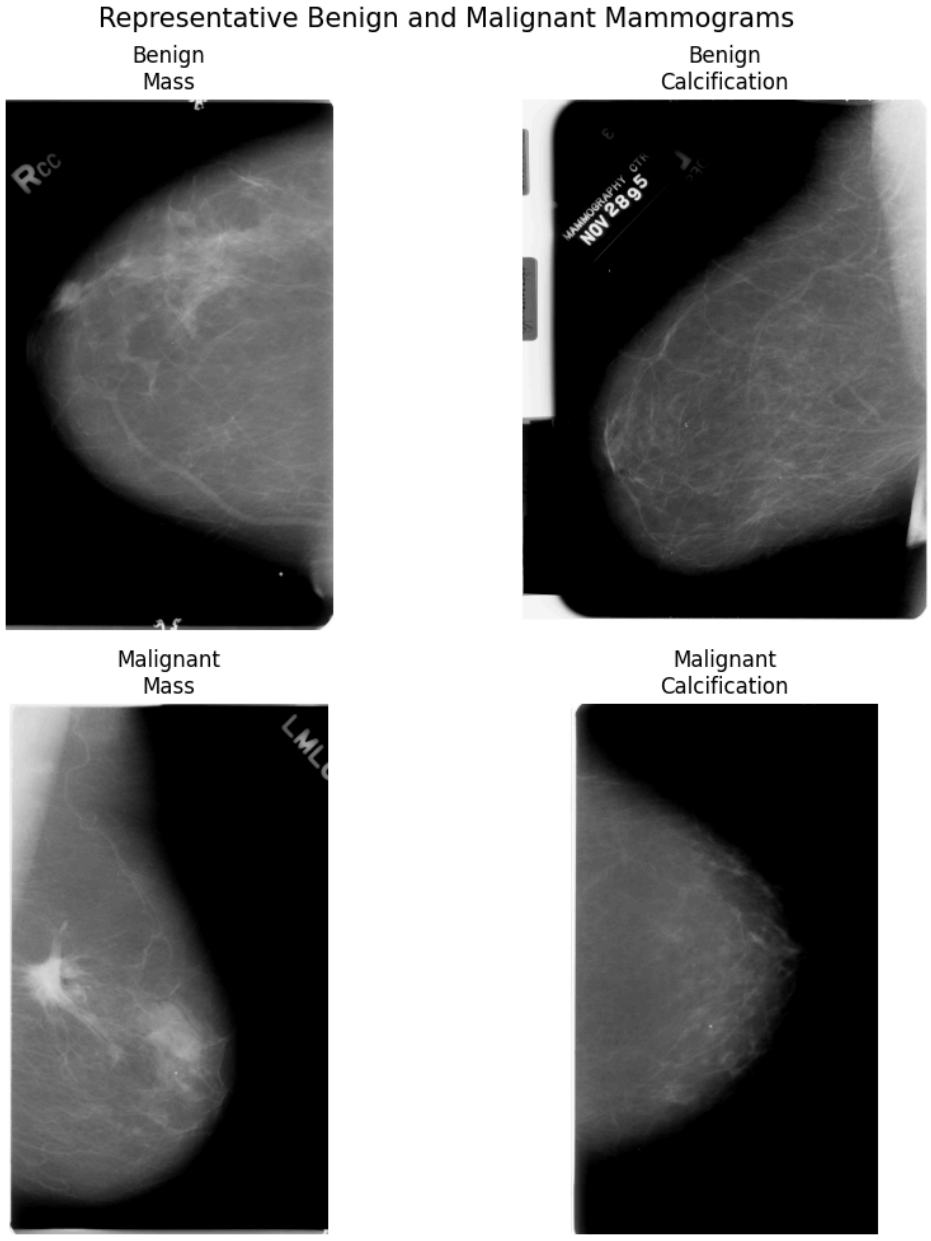


Figure 4.1. Representative benign and malignant full mammograms from the CBIS-DDSM dataset, including mass and calcification cases.

Table 4.1. Summary of the raw metadata and final matched dataset.

Item	Number
Combined case-description records	3568
Benign case-description records	2111
Malignant case-description records	1457
Cropped image records	3567
ROI mask records	3247
Full Mammography records in image metadata	2857
Final matched lesion linked records	3286
Benign matched records	1931
Malignant matched records	1355
Mass records	1696
Calcification records	1590
MLO-view records	1743
CC-view records	1543
Unique patients	1460

The final matched dataset was relatively balanced by lesion type, containing 1,696 mass and 1,590 calcification records. It contained 1,743 mediolateral oblique records and 1,543 craniocaudal records. Therefore, neither lesion type nor mammographic view overwhelmingly dominated the modelling data. However, the moderate benign-class majority justified the use of class weighting during training and evaluation using metrics beyond accuracy.

Exploration of the processed inputs confirmed that every image was resized to 224 × 224 pixels and converted into three-channel format. The three-channel conversion duplicated the grayscale information to satisfy the expected model input shape rather than adding new visual information. Resizing standardised the tensor dimensions and reduced computational requirements, but comparison with the original mammograms showed that fine anatomical detail became less visible at the lower resolution.

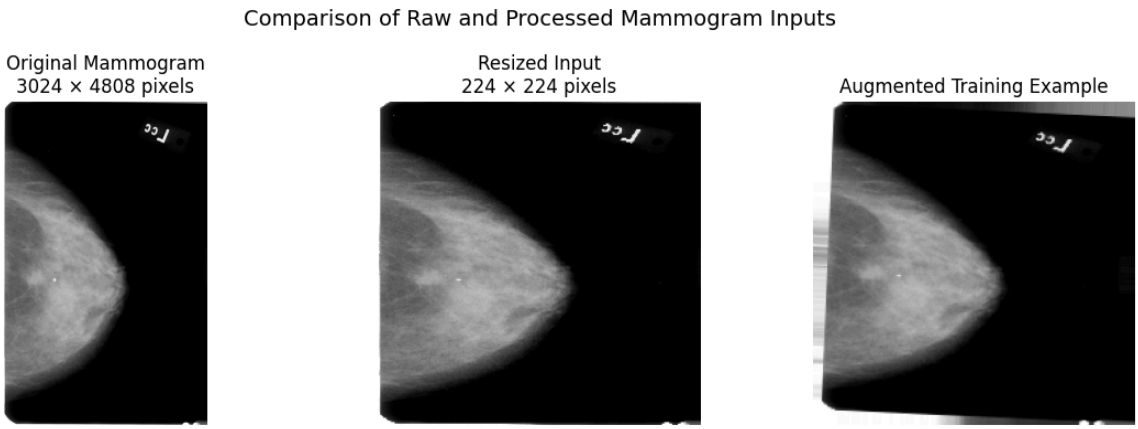


Figure 4.2. Comparison of an original full mammogram, its resized 224 x 224 model input, and an augmented training example.

Figure 4.2 shows how preprocessing transformed an original 3024 x 4808 pixel mammogram into the 224 x 224 input required by the models. Resizing standardised the input dimensions and reduced computational requirements, but it also altered the original aspect ratio and reduced the visibility of fine image details. The augmented example demonstrates the moderate geometric variation introduced during training. The surrounding background and embedded acquisition marker remained visible after preprocessing, meaning that the models could potentially learn from non-clinical image features in addition to breast tissue.

4.3 Dataset Splitting and Preprocessing

The dataset was divided by patient rather than by individual image to reduce the risk of data leakage. A patient was assigned a malignant patient-level label if any associated mammogram was malignant. This produced 752 benign and 708 malignant patients.

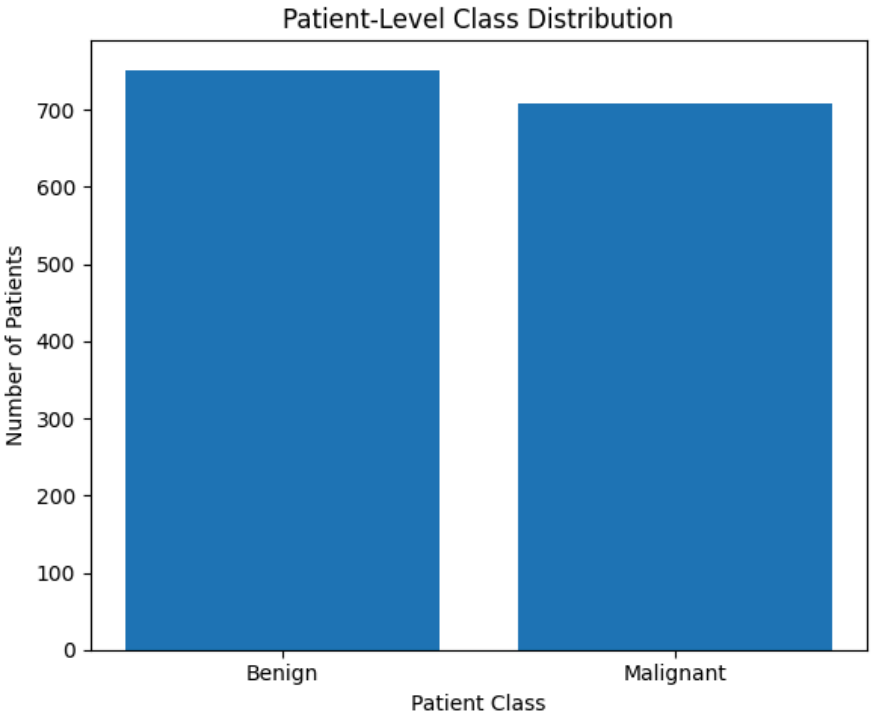


Figure 4.3. Patient-level class distribution before stratified dataset splitting.

The patients were divided into approximately 70% training, 15% validation, and 15% testing partitions using stratified splitting. All images belonging to one patient remained within the same partition.

Table 4.2. Patient-Level Dataset Split.

Partition	**Patients**	**Records**	**Benign Records**	**Malignant Records**	**Benign (%)**	**Malignant (%)**
Training	1,022	2,289	1,346	943	58.80	41.20
Validation	219	529	316	213	59.74	40.26
Testing	219	468	269	199	57.48	42.52
Total	**1460**	**3286**	**1931**	**1355**	-	-

Patient-overlap checks returned zero shared patients between every pair of partitions. Therefore, images belonging to one patient did not appear across the training, validation, or testing data.

All images were resized to 224 × 224 pixels and converted into three-channel format. Pixel values were rescaled to the range 0 - 1 for the custom CNNs. ResNet50 and

DenseNet121 used their corresponding preprocess_input functions, while EfficientNetB0 used the input rescaling incorporated into its Keras implementation.

Training augmentation consisted of rotation up to 10 degrees, horizontal flipping, and zooming up to 10%. Augmentation was applied only to the training generators, while validation and testing images remained unchanged.

The training partition contained 1,346 benign and 943 malignant images. Class weights were calculated using the training labels, producing weights of approximately 0.850 for benign cases and 1.214 for malignant cases. The higher malignant weight increased the contribution of errors involving the less represented malignant class without discarding benign training samples.

4.4 Custom CNN Development

Three custom CNN configurations were implemented to investigate whether increasing convolutional depth improved classification performance.

The three-block CNN served as the original baseline. It contained convolutional blocks with 32, 64, and 128 filters. Each block used a 3 × 3 convolution with ReLU activation, Batch Normalisation, and Max Pooling. The resulting feature maps were flattened and passed to a 128-neuron Dense layer, followed by Dropout of 0.5 and a sigmoid output neuron.

Code Extract 4.1. Three-block baseline CNN architecture.

```
baseline_cnn = Sequential([
    Input(shape=(224, 224, 3)),

    Conv2D(32, (3, 3), activation='relu'),
    BatchNormalization(),
    MaxPooling2D(2, 2),

    Conv2D(64, (3, 3), activation='relu'),
    BatchNormalization(),
    MaxPooling2D(2, 2),

    Conv2D(128, (3, 3), activation='relu'),
    BatchNormalization(),
    MaxPooling2D(2, 2),
```

```
    Flatten(),

    Dense(128, activation='relu'),
    Dropout(0.5),

    Dense(1, activation='sigmoid')
])
```

A shallower two-block CNN used 32 and 64 filters, while the deeper four-block CNN added a fourth convolutional block containing 256 filters. The same Dense, Dropout, and sigmoid classification structure was used across the three configurations.

Table 4.3. Custom CNN configurations.

Configuration	Convolutional filters	Total parameters	Purpose
Two-block CNN	32 → 64	23,907,905	Investigate a shallower feature extractor
Three-block CNN	32 → 64 → 128	11,169,985	Provide the original baseline
Four-block CNN	32 → 64 → 128 → 256	5,109,185	Investigate greater convolutional depth

The two-block CNN had the largest parameter count despite containing the fewest convolutional blocks. This occurred because fewer pooling operations produced a larger feature map before the Flatten layer, resulting in approximately 23.89 million parameters in its Dense layer. Conversely, additional pooling in the four-block CNN reduced the flattened feature size and therefore lowered its overall parameter count. Convolutional depth and total parameter count consequently did not increase together.

All configurations were trained using Adam, binary cross-entropy, class weighting, EarlyStopping, ModelCheckpoint, and ReduceLROnPlateau. Accuracy, precision, recall, and AUC were monitored during training.

4.5 Frozen Transfer-Learning Implementation

ResNet50, EfficientNetB0, and DenseNet121 were loaded with ImageNet-pretrained weights and without their original classification heads. During the initial transfer-learning stage, each convolutional backbone was frozen.

A common classification head was attached to each backbone. Global Average Pooling converted the final feature maps into a feature vector, Dropout of 0.5 provided regularisation, and a sigmoid neuron generated the probability of a malignant case.

Code Extract 4.2. Common transfer-learning classification structure.

```
resnet_model = tf.keras.Sequential([
    resnet_base,

    tf.keras.layers.GlobalAveragePooling2D(),

    tf.keras.layers.Dropout(0.5),

    tf.keras.layers.Dense(
        1,
        activation='sigmoid'
    )
])
```

Global Average Pooling avoided the large parameter increase that would have resulted from flattening the complete feature maps. The same classification-head structure and training settings were used for all three pretrained architectures.

Table 4.4. Frozen transfer-learning configurations.

Model	Base Output	Total Parameters	Trainable Parameters	Non-Trainable Parameters
ResNet50	7 × 7 × 2,048	23,589,761	2,049	23,587,712
EfficientNetB0	7 × 7 × 1,280	4,050,852	1,281	4,049,571
DenseNet121	7 × 7 × 1,024	7,038,529	1,025	7,037,504

During the frozen stage, only the newly added classification layer contained trainable parameters. This allowed each architecture to be evaluated initially as a fixed pretrained feature extractor.

4.6 Model Training and Fine-Tuning

All models used the same patient-level partitions and class weights. Initial training used Adam with a learning rate of $1\ X\ 10^{-4}$, binary cross-entropy loss, and a batch size of 32.

EarlyStopping monitored validation loss with a patience of five epochs and restored the weights from the best epoch. ModelCheckpoint saved the model producing the lowest validation loss. ReduceLROnPlateau reduced the learning rate by a factor of 0.2 after three epochs without improvement.

Table 4.5. Model Training Configuration.

Configuration	Custom CNNs	Transfer Learning models
Optimiser	Adam	Adam
Initial learning rate	1×10^{-4}	1×10^{-4}
Loss function	Binary cross-entropy	Binary cross-entropy
Batch size	32	32
EarlyStopping	Yes	Yes
ReduceLROnPlateau	Yes	Yes
Class weighting	Yes	Yes
ModelCheckpoint	Yes	Yes

Following frozen-backbone training, all three pretrained models were fine-tuned. Only selected upper regions of each backbone were made trainable, while earlier layers remained frozen. Batch Normalisation layers also remained frozen to preserve their pretrained statistics.

The models were recompiled using a lower learning rate of (1 \times 10^{-5}). This allowed higher-level features to adapt gradually to mammography images while reducing the risk of large changes to the pretrained weights.

Table 4.6. Fine-tuning configurations.

Model	Fine-Tuned Region	Trainable backbone layers after Batch Normalisation freezing	Trainable parameters

ResNet50	Final conv5 stage	22	14,955,521
EfficientNetB0	Final 30 layers	23	1,484,641
DenseNet121	Final 40 layers	29	816,129

The difference between the number of layers considered and the number remaining trainable for EfficientNetB0 and DenseNet121 resulted from keeping Batch Normalisation layers frozen. The best weights from each frozen and fine-tuned training stage were retained for evaluation.

4.7 Model Evaluation and Selection

After training, each model generated malignant-class probabilities for the 468 record test partition. Initial predictions were obtained using a classification threshold of 0.50.

Performance was calculated using accuracy, malignant precision, malignant recall or sensitivity, specificity, F1-score, AUC, ROC curves, and confusion matrices. False-positive and false-negative counts were also recorded to identify differences in the types of errors produced by the models.

Frozen and fine-tuned versions of the pretrained models were compared under the same test conditions. Fine-tuned ResNet50 produced the strongest overall combination of accuracy, malignant recall, F1-score, and AUC at the default threshold and was therefore selected for additional threshold and error analysis. The detailed comparative results and their interpretation are presented in Chapter 5.

4.8 Validation-Based Threshold Selection

The fine-tuned ResNet50 initially used the default classification threshold of 0.50. However, because malignant-case detection was an important consideration, threshold analysis was performed to investigate the trade-off between sensitivity and specificity.

The fine-tuned model generated probabilities for the 529 validation records. Candidate thresholds from 0.30 to 0.70 were evaluated in increments of 0.05. At every threshold, accuracy, malignant precision, sensitivity, specificity, F1-score, false positives, and false negatives were calculated.

Youden's J statistic was also calculated:

$$J = Sensitivity + Specificity - 1$$

A threshold of 0.40 produced the highest Youden's J statistic on the validation partition and was therefore selected. The threshold was fixed before it was applied to the test probabilities. Threshold selection itself was performed without reference to the test labels.

Chapter 5 compares the selected threshold of 0.40 with the default threshold of 0.50 and examines the resulting changes in malignant recall, specificity, false positives, and false negatives.

4.9 Classification-Error Analysis

After applying the selected threshold, test predictions were separated into true-positive, true-negative, false-positive, and false-negative categories. The corresponding image paths, true labels, predicted labels, and malignant probabilities were recorded.

Representative errors were selected according to their predicted probabilities. False-positive examples included cases close to the threshold, cases with moderate malignant probabilities, and highly confident incorrect malignant predictions. False-negative examples similarly included cases close to the threshold and highly confident incorrect benign predictions.

This selection method distinguished borderline errors from confident errors. Confident false negatives were particularly important because their probabilities were substantially below the selected threshold, meaning that a small threshold adjustment alone would not correct them.

4.10 Grad-CAM Implementation

Grad-CAM was implemented for the final fine-tuned ResNet50 using conv5_block3_out, the final convolutional layer in its backbone. For each image, the gradient of the malignant-class output was calculated with respect to the convolutional feature maps. The gradients were spatially averaged and used to weight the feature maps. Positive contributions were retained and normalised to produce a heatmap between zero and one.

The heatmaps were resized and overlaid on the corresponding mammograms. Representative examples were generated for four prediction categories:

- The true positive with the highest malignant probability.
- The true negative with the lowest malignant probability.
- The false positive with the highest malignant probability.
- The false negative with the lowest malignant probability.

These selections provided examples of confident correct and confident incorrect predictions. Grad-CAM was used as an exploratory explanation method rather than evidence of lesion localisation. Because the heatmaps were not quantitatively compared with lesion-level masks or bounding boxes, they could not confirm whether the highlighted regions corresponded to the true abnormalities.

4.11 Implementation Summary

The implementation produced an end-to-end pipeline using 3,286 lesion-linked full-mammogram records. Patient-level splitting prevented overlap between partitions, while augmentation and class weighting addressed training variation and class imbalance. Three custom CNNs and frozen and fine-tuned versions of three pretrained architectures were implemented. Fine-tuned ResNet50 was selected for threshold analysis, error analysis, and Grad-CAM, with the results evaluated in Chapter 5.

CHAPTER 5: EVALUATION (2493/2500 words)

5.1 Evaluation Overview

This chapter compares three custom CNN configurations and frozen and fine-tuned versions of ResNet50, EfficientNetB0, and DenseNet121 using the same held-out test partition of 468 full-mammogram records from 219 patients. Evaluation used accuracy, malignant precision, sensitivity, specificity, F1-score, AUC, confusion matrices, and false-positive and false-negative counts. Fine-tuned ResNet50 was subsequently examined through validation-based threshold selection, error analysis, and Grad-CAM.

5.2 Evaluation Methodology

Each model generated a malignant-class probability for every test record. Initial predictions used the default threshold of 0.50, with malignant cases treated as the positive class. Accuracy measured overall correctness; precision measured the proportion of malignant predictions that were correct; sensitivity and specificity measured correct malignant and benign classification respectively; F1-score combined precision and sensitivity; and AUC assessed discrimination across thresholds.

Fine-tuned ResNet50 probabilities from the validation partition were then used to select an alternative threshold, which was applied to the test probabilities and compared with the default threshold.

5.3 Custom CNN Scaling Experiment

Table 5.1 presents the results of the two-, three-, and four-block custom CNNs. These models used the same dataset partitions, augmentation, class weighting, optimiser, and training callbacks.

Table 5.1. Test performance of the custom CNN configurations.

CNN configuration	Accuracy	Malignant precision	Sensitivity	Specificity	Malignant F1	AUC	False Positive	False Negative
Two-block CNN	0.6474	0.5702	0.6935	0.6134	0.6259	0.7179	104	61
Three-block CNN	0.6197	0.5425	0.6734	0.5799	0.6009	0.6688	113	65
Four-block CNN	0.6368	0.5612	0.6683	0.6134	0.6101	0.7025	104	66

The original three-block CNN served as the experimental benchmark. It achieved an accuracy of 0.6197, malignant precision of 0.5425, sensitivity of 0.6734, specificity of 0.5799, F1-score of 0.6009, and AUC of 0.6688, with 113 false positives and 65 false negatives.

The two-block CNN achieved the strongest custom-CNN performance, including the highest accuracy, sensitivity, F1-score, and AUC and the fewest false negatives. The four-block model improved on the benchmark but did not outperform the two-block configuration.

Increasing convolutional depth therefore did not consistently improve generalisation. However, the two-block model contained greater fully connected capacity because its larger feature map was flattened before the Dense layer. Its advantage cannot be attributed solely to reduced depth; the results show only that the tested four-block expansion provided no improvement under the selected configuration.

5.4 Frozen and Fine-Tuned Transfer Learning

Table 5.2 compares each pretrained architecture before and after fine-tuning. All results in this table use the default threshold of 0.50.

Table 5.2. Frozen and fine-tuned transfer learning performance.

Model	Accuracy	Malignant Precision	Sensitivity	Specificity	Malignant F1	AUC	FP	FN
ResNet50 frozen	0.6453	0.5892	0.5477	0.7175	0.5677	0.7337	76	90
ResNet50 fine-tuned	0.7009	0.6398	0.6784	0.7175	0.6585	0.7846	76	64
EfficientNetB0 frozen	0.6560	0.6397	0.4372	0.8178	0.5194	0.7125	49	112
EfficientNetB0 fine-tuned	0.6645	0.6346	0.4975	0.7881	0.5577	0.7499	57	100
DenseNet121 frozen	0.6581	0.5951	0.6131	0.6914	0.6040	0.7103	83	77
DenseNet121 fine-tuned	0.6816	0.6812	0.4724	0.8364	0.5579	0.7469	44	105

Fine-tuning improved the AUC of all three pretrained architectures, suggesting that adapting higher-level ImageNet features improved probability-based discrimination.

However, the effect on threshold-dependent performance differed substantially between models.

ResNet50 showed the greatest and most consistent improvement. Its accuracy increased by 5.56 percentage points, sensitivity increased from 0.5477 to 0.6784, and AUC increased from 0.7337 to 0.7846. False negatives decreased from 90 to 64 without increasing false positives.

EfficientNetB0 showed smaller improvements. Fine-tuning increased sensitivity from 0.4372 to 0.4975 and reduced false negatives from 112 to 100. Its AUC increased to 0.7499, although false positives increased from 49 to 57.

DenseNet121 demonstrated a different trade-off. Fine-tuning increased accuracy, malignant precision, specificity, and AUC and reduced false positives from 83 to 44. However, sensitivity decreased from 0.6131 to 0.4724, increasing false negatives from 77 to 105. Fine-tuning therefore made DenseNet121 more conservative in predicting malignant cases.

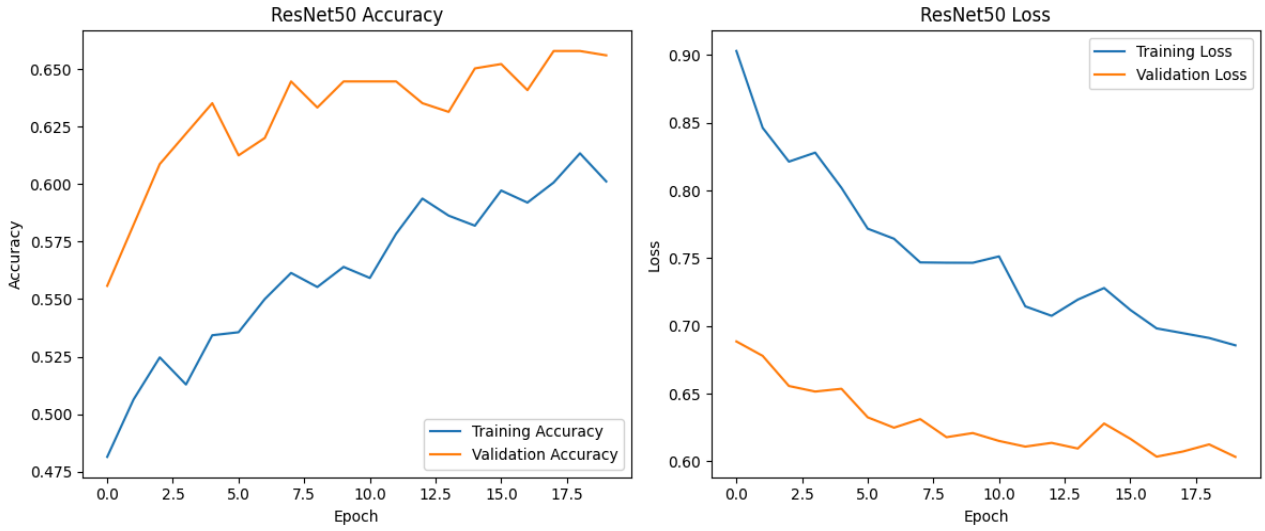


Figure 5.1. Training and validation accuracy and loss for fine-tuned ResNet50.

The fine-tuned ResNet50 training curves showed continuing improvement in training performance, while validation performance fluctuated and plateaued earlier, indicating some divergence. Early stopping and restoration of the best validation-loss weights limited later overfitting, although performance remained sensitive to the selected training stage.

5.5 Comparative Model Evaluation

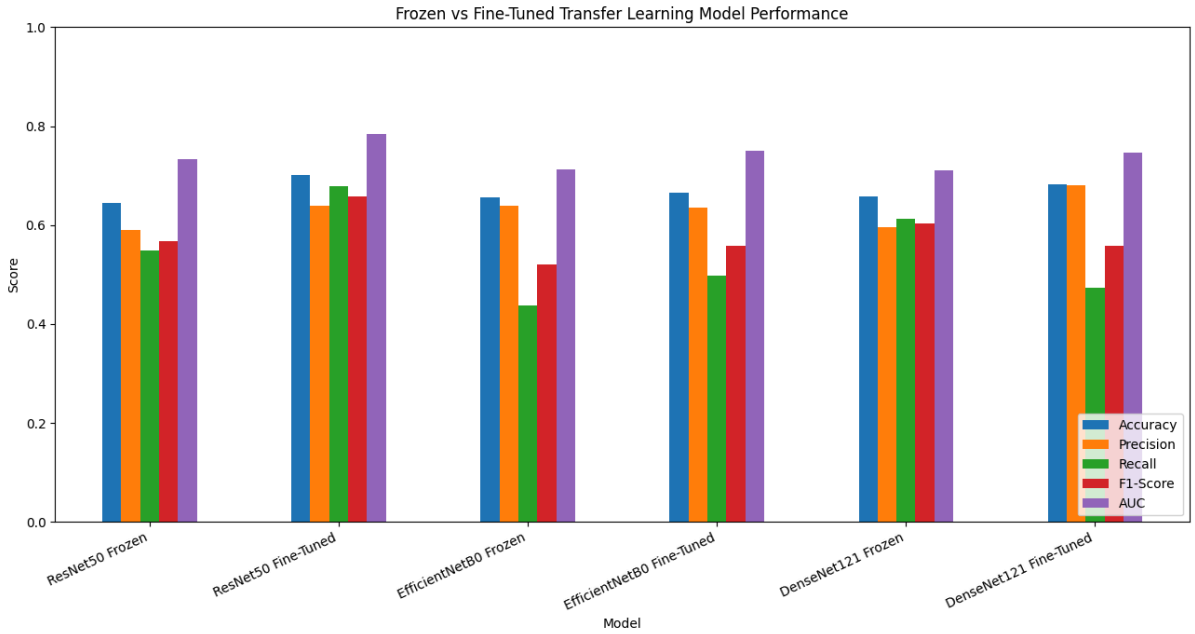


Figure 5.2. Comparison of accuracy, malignant precision, sensitivity, F1-score, and AUC across frozen and fine-tuned transfer-learning models.

No architecture achieved the best result for every metric. Fine-tuned ResNet50 achieved the highest accuracy, sensitivity, F1-score, and AUC among the transfer-learning configurations. Fine-tuned DenseNet121 achieved the highest malignant precision and specificity but substantially lower sensitivity.

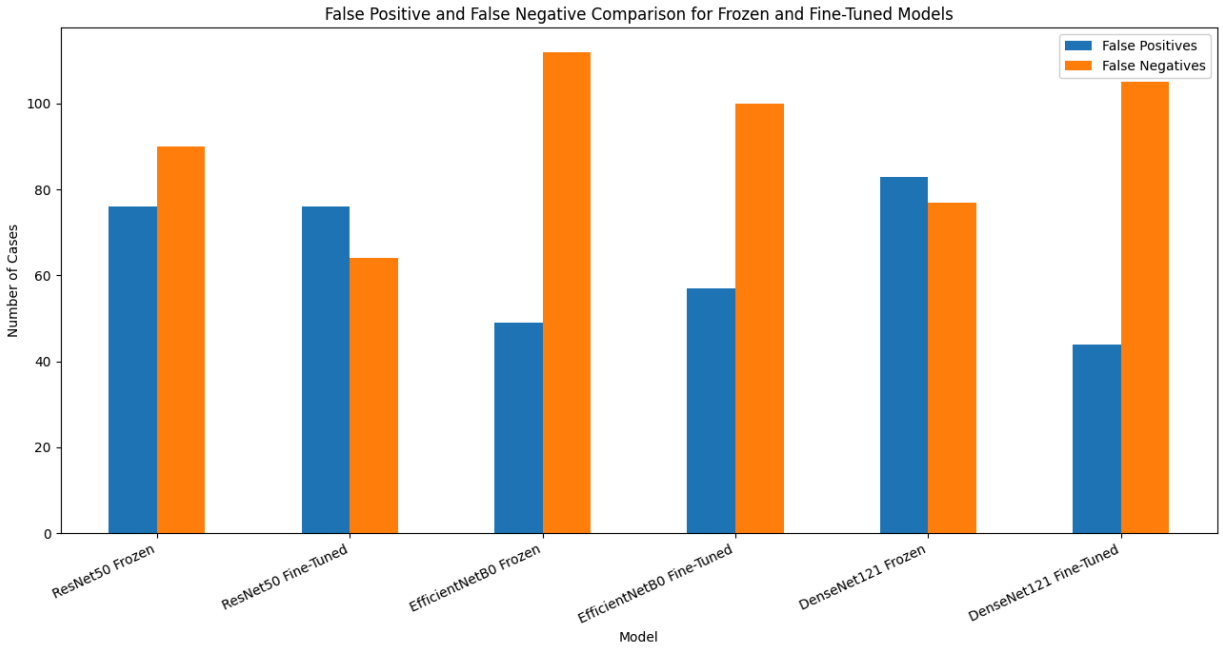


Figure 5.3. Comparison of false-positive and false-negative counts across frozen and fine-tuned transfer-learning models.

Figure 5.3 further illustrates the different error profiles. Fine-tuned DenseNet121 produced only 44 false positives but missed 105 malignant images. In comparison, fine-tuned ResNet50 produced 76 false positives and 64 false negatives. Therefore, the lower false-positive count of DenseNet121 did not represent stronger malignant-case detection.

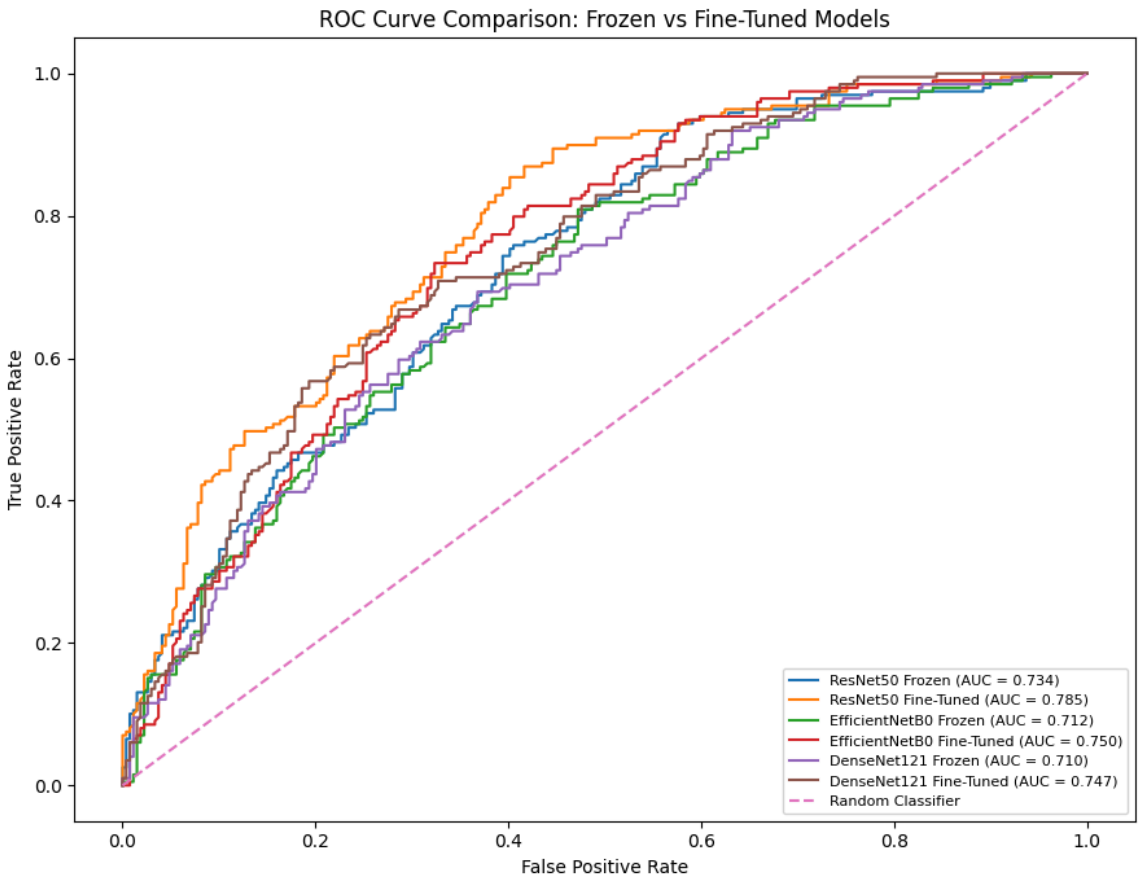


Figure 5.4. ROC curves for the frozen and fine-tuned transfer-learning models.

Fine-tuned ResNet50 achieved the highest AUC of 0.7846. This indicates that it provided the strongest overall ranking of malignant cases above benign cases across possible thresholds. Its combination of discrimination, accuracy, sensitivity, and F1-score supported its selection for further threshold analysis.

The two-block custom CNN remained competitive, achieving an AUC of 0.7179 and sensitivity of 0.6935. Its sensitivity was slightly higher than that of fine-tuned ResNet50 at the default threshold, although its accuracy, precision, F1-score, and AUC were lower. This demonstrates that transfer learning did not automatically dominate every metric but provided the strongest overall result after fine-tuning.

Relative to the three-block CNN benchmark, fine-tuned ResNet50 at the default threshold improved accuracy from 0.6197 to 0.7009, representing an increase of 8.12 percentage points. Malignant precision increased from 0.5425 to 0.6398, specificity increased from 0.5799 to 0.7175, F1-score increased from 0.6009 to 0.6585, and AUC increased from 0.6688 to 0.7846. Sensitivity improved only slightly from 0.6734 to 0.6784, but false positives decreased from 113 to 76 and false negatives decreased from 65 to 64.

Fine-tuned ResNet50 therefore demonstrated a clear improvement over the benchmark in overall discrimination and classification balance, although its default-threshold improvement in malignant sensitivity was limited.

5.6 Validation-Based Threshold Selection

Thresholds from 0.30 to 0.70 were evaluated using the fine-tuned ResNet50 predictions on the validation partition. Table 5.3 presents sensitivity, specificity, F1-score, and Youden's J for each threshold.

Table 5.3. Fine-tuned ResNet50 threshold performance on the validation partition.

Threshold	Recall/Sensitivity	Specificity	F1-score	Youden's J
0.30	0.9343	0.4209	0.6689	0.3552
0.35	0.9108	0.4684	0.6748	0.3792
0.40	0.8545	0.5316	0.6703	0.3861
0.45	0.7840	0.5886	0.6549	0.3726
0.50	0.7324	0.6361	0.6446	0.3685
0.55	0.6808	0.6709	0.6277	0.3516
0.60	0.6197	0.7310	0.6140	0.3507
0.65	0.5164	0.7911	0.5656	0.3076
0.70	0.4601	0.8576	0.5506	0.3177

A threshold of 0.35 produced the highest validation F1-score, while 0.40 produced the highest Youden's J. The threshold of 0.40 was selected because it provided the best observed balance between sensitivity and specificity according to the predefined Youden's J criterion.

Table 5.4 compares the selected threshold with the default threshold on the test partition.

Table 5.4. Fine-tuned ResNet50 test performance at thresholds 0.50 and 0.40.

Threshold	Accuracy	Malignant precision	Sensitivity	Specificity	F1-score	False positives	False Negatives
Default 0.50	0.7009	0.6398	0.6784	0.7175	0.6585	76	64

Selected 0.40	0.6987	0.6124	0.7940	0.6283	0.6915	100	41

Lowering the threshold increased sensitivity from 0.6784 to 0.7940 and reduced false negatives from 64 to 41. The malignant F1-score also increased from 0.6585 to 0.6915. Accuracy decreased only slightly, from 0.7009 to 0.6987.

However, the improvement in malignant detection was accompanied by an increase in false positives from 76 to 100, while specificity decreased from 0.7175 to 0.6283. This demonstrates that threshold optimisation did not improve every aspect of performance; it shifted the operating point towards greater sensitivity.

Compared with the original three-block CNN benchmark, fine-tuned ResNet50 at the selected threshold of 0.40 increased sensitivity from 0.6734 to 0.7940 and F1-score from 0.6009 to 0.6915. It also reduced false negatives from 65 to 41 and false positives from 113 to 100. These results provide quantitative evidence that the final sensitivity-focused configuration improved malignant-case detection relative to the original benchmark.

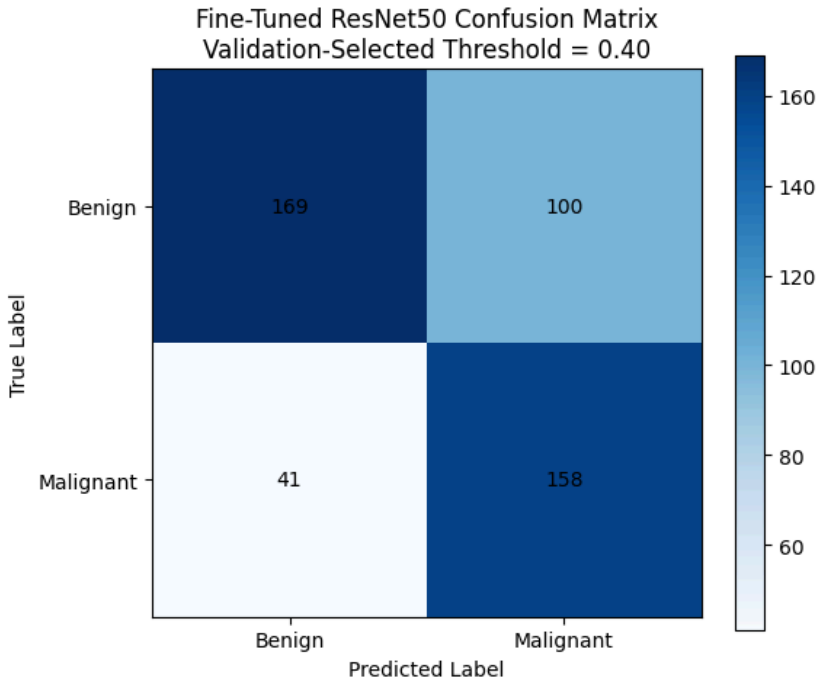


Figure 5.5. Confusion matrix for fine-tuned ResNet50 at the validation-selected threshold of 0.40.

At the selected threshold, the model correctly identified 158 of the 199 malignant images and 169 of the 269 benign images. The AUC remained 0.7846 because changing the classification threshold does not alter the underlying predicted probabilities.

5.7 False-Positive and False-Negative Analysis

The final thresholded ResNet50 produced 100 false positives and 41 false negatives. Representative errors were selected at different probability levels to distinguish borderline errors from confident errors.

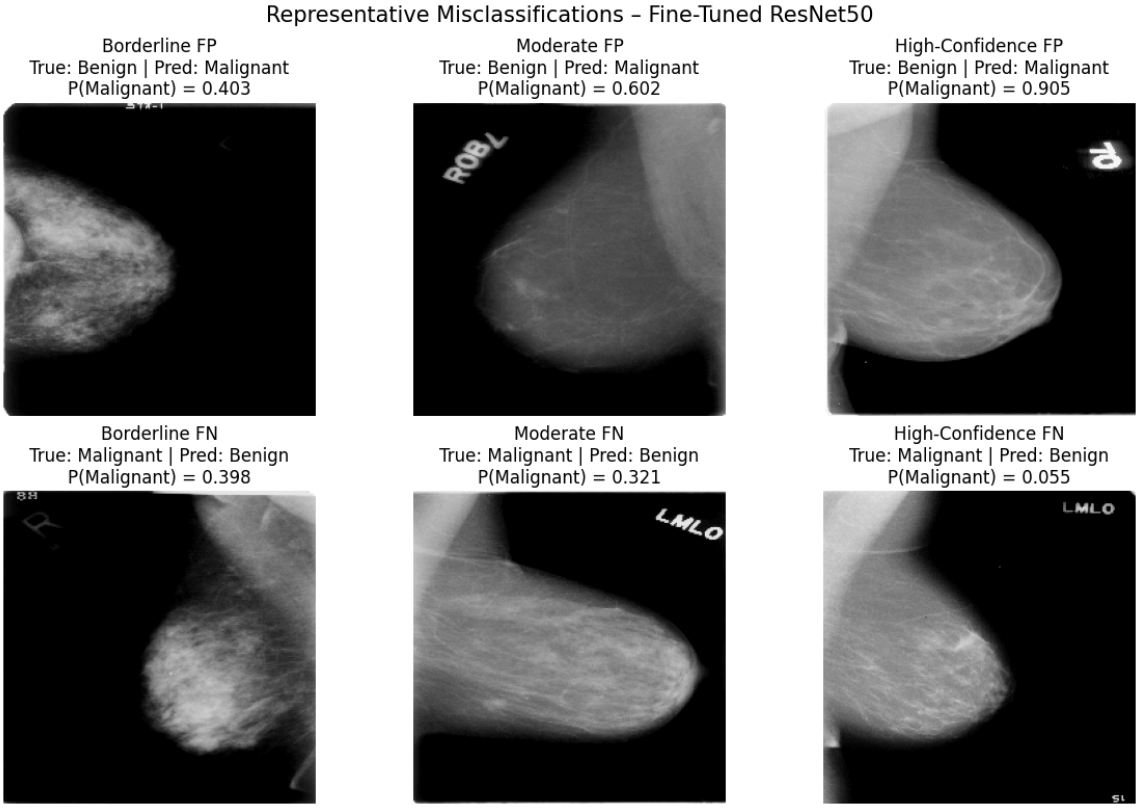


Figure 5.6. Representative borderline, moderate, and high-confidence false-positive and false-negative predictions from fine-tuned ResNet50 at the selected threshold of 0.40.

The selected false-positive examples had malignant probabilities of 0.4028, 0.6023, and 0.9050. The first prediction was close to the threshold and could change with a small threshold adjustment. In contrast, the probability of 0.9050 represented a confident error in which a benign image was assigned a high malignant probability.

The selected false-negative examples had malignant probabilities of 0.3980, 0.3210, and 0.0548. The first was close to the threshold, while the probability of 0.0548 represented a confident failure to identify a malignant image. Such confident errors are unlikely to be corrected through a minor threshold adjustment alone and may reflect limitations in the

learned features, image resolution, or the presence of visual characteristics not captured by the model.

Visual inspection alone could not establish the exact clinical cause of each error. The representative cases therefore illustrate different levels of prediction confidence rather than providing a definitive explanation of misclassification.

5.8 Grad-CAM Analysis

Grad-CAM visualisations were generated for a representative true positive, true negative, false positive, and false negative from the final fine-tuned ResNet50. The selected examples had malignant probabilities of 0.9917, 0.0111, 0.9050, and 0.0548 respectively.

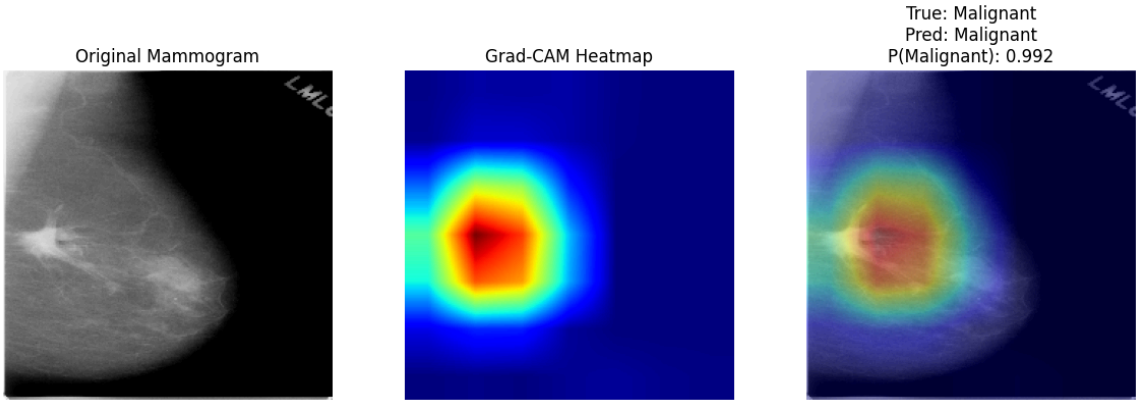


Figure 5.7. Grad-CAM visualisations for representative true-positive prediction.

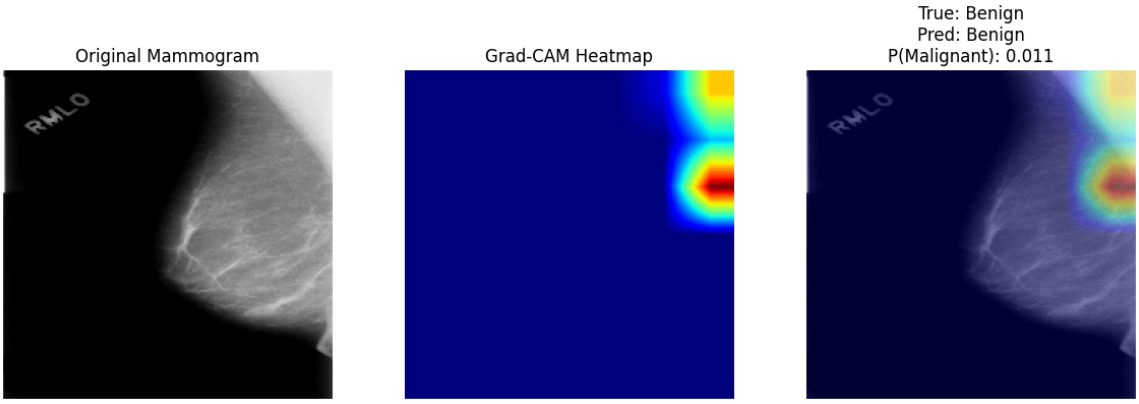


Figure 5.8. Grad-CAM visualisations for representative true-negative prediction.

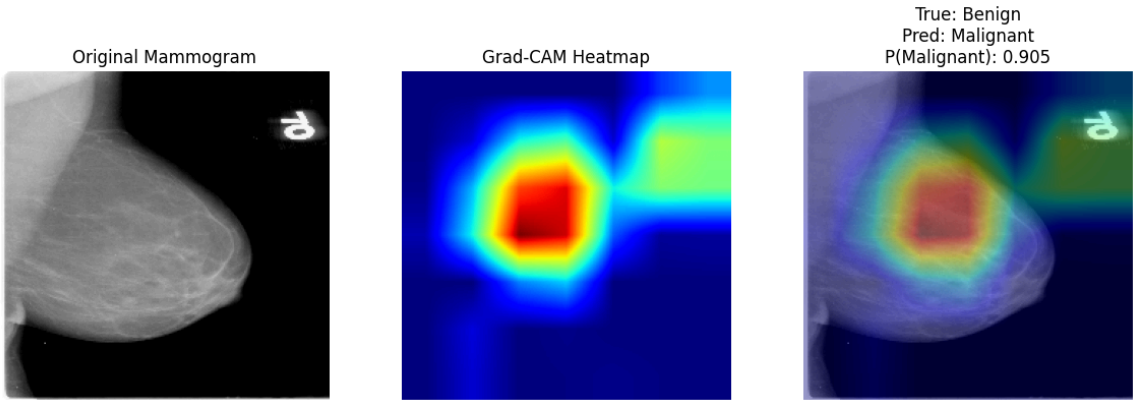


Figure 5.9. Grad-CAM visualisations for representative false-positive prediction.

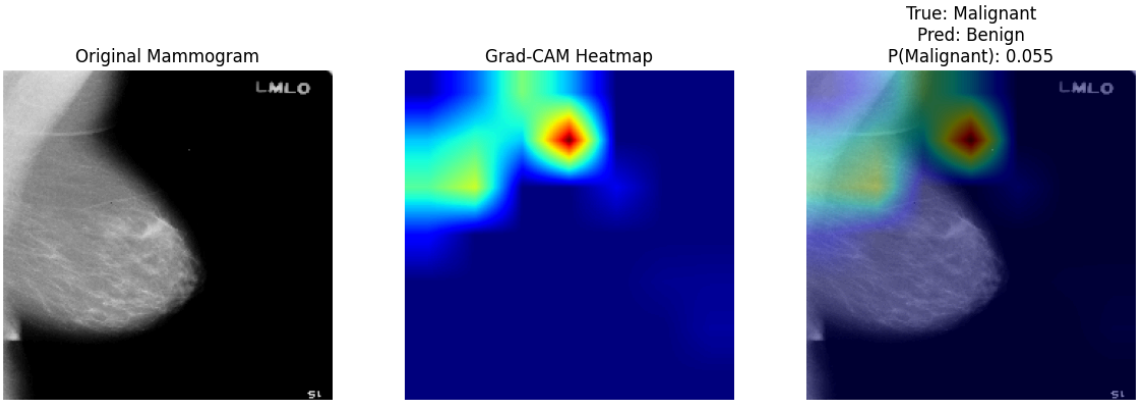


Figure 5.10. Grad-CAM visualisations for representative false-negative prediction.

Some heatmaps showed strong activation within the breast region, suggesting that the model used mammographic image content when producing its predictions. However, activation was also observed near image borders and embedded view markers in some cases. This indicates that the model may occasionally rely on acquisition-related or non-clinical image features.

The false-positive and false-negative examples demonstrate that strong activation does not necessarily correspond to a correct prediction. Grad-CAM therefore provides exploratory insight into model behaviour but does not prove that the model identified a clinically relevant lesion. The heatmaps were not compared quantitatively with lesion masks or bounding boxes, so localisation accuracy could not be established.

5.9 Critical Evaluation and Limitations

The final fine-tuned ResNet50 achieved stronger performance than the models reported in the draft evaluation. Its AUC of 0.7846 indicated moderate discrimination, while

threshold adjustment increased malignant sensitivity to 0.7940 and reduced false negatives to 41. Nevertheless, the results remain insufficient for clinical deployment.

One limitation is the resizing of high-resolution mammograms to 224 × 224 pixels. This substantially reduces computational requirements but may remove subtle masses, calcifications, margins, and texture patterns that are relevant to classification. A higher-resolution or crop-based approach may preserve more clinically relevant detail.

The evaluation used one patient-level split and one completed training run for each configuration. Neural-network training is affected by random initialisation, minibatch order, augmentation, and optimisation. Without repeated runs or patient-level bootstrap confidence intervals, small differences between models should not be treated as definitive.

Although the data were separated by patient, predictions and metrics were calculated at record level. Lesion-linked matching could repeat a full mammogram when multiple abnormalities were annotated, giving some mammograms additional weight, while patients could also contribute multiple views. Future work should construct one record per unique mammogram using an explicit pathology-aggregation rule, before aggregating predictions at breast or patient level and applying patient-level resampling.

Fine-tuning improved AUC for every pretrained architecture but did not consistently improve malignant sensitivity. The DenseNet121 result demonstrates that higher accuracy or specificity can coincide with more false negatives. Fine-tuning strategies may therefore require architecture-specific layer selection and validation rather than identical assumptions across networks.

The selected threshold maximised Youden's J, which gives equal importance to sensitivity and specificity. A clinically oriented system might instead require a predefined minimum sensitivity and then select the threshold providing the greatest achievable specificity. Probability calibration should also be examined using reliability curves and measures such as the Brier score.

The threshold was selected using validation data, but ResNet50 was chosen for further analysis after the comparative test evaluation. The final thresholded result should therefore be interpreted as an exploratory extension rather than a completely untouched confirmatory test estimate. A stronger future design would select both the model and threshold using validation data before evaluating the complete pipeline once on the test partition.

External validation was not conducted. CBIS-DDSM may differ from contemporary screening data in acquisition equipment, population characteristics, processing, and image quality. Performance on this dataset therefore cannot establish generalisation to other institutions or clinical environments.

Finally, Grad-CAM does not provide causal or clinical explanations. Without comparison against lesion annotations and assessment by qualified imaging specialists, the highlighted regions cannot confirm medically appropriate reasoning.

5.10 Evaluation Summary

The two-block CNN achieved the strongest custom-CNN performance, while fine-tuning increased AUC across all three pretrained architectures but affected sensitivity differently. At the default threshold, fine-tuned ResNet50 achieved the strongest overall transfer-learning result, with an accuracy of 0.7009 and AUC of 0.7846.

Applying the validation-selected threshold of 0.40 increased sensitivity from 0.6784 to 0.7940 and reduced false negatives from 64 to 41, although false positives increased and specificity decreased. The findings therefore show that architecture, fine-tuning, and threshold selection influenced performance differently. Fine-tuned ResNet50 at 0.40 provided the strongest sensitivity-focused configuration, but further validation is required.

CHAPTER 6:CONCLUSION (659/1000 words)

6.1 Project Summary

This project developed and evaluated deep learning approaches for classifying CBIS-DDSM full-mammogram records as benign or malignant. Three custom CNN configurations were compared with frozen and fine-tuned versions of ResNet50, EfficientNetB0, and DenseNet121 using common patient-level partitions.

Models were evaluated using accuracy, malignant precision, sensitivity, specificity, F1-score, AUC, and error counts. Fine-tuned ResNet50 was investigated further through validation-based threshold selection, representative error analysis, and Grad-CAM, extending the project beyond a direct architectural comparison.

6.2 Key Findings

The two-block CNN achieved the strongest custom-CNN performance, with an accuracy of 0.6474, sensitivity of 0.6935, F1-score of 0.6259, and AUC of 0.7179. However, its greater fully connected capacity means that its advantage cannot be attributed solely to reduced convolutional depth.

Fine-tuning increased AUC for all pretrained architectures but affected their threshold-dependent performance differently. Fine-tuned ResNet50 provided the strongest overall default-threshold result, achieving an accuracy of 0.7009, sensitivity of 0.6784, and AUC of 0.7846. EfficientNetB0 reached an AUC of 0.7499 but lower sensitivity, while DenseNet121 achieved the highest precision and specificity but produced 105 false negatives. Therefore, higher precision or specificity did not necessarily indicate stronger malignant-case detection.

The validation-selected threshold of 0.40 increased ResNet50 sensitivity to 0.7940, improved malignant F1-score to 0.6915, and reduced false negatives from 64 to 41. Accuracy remained similar at 0.6987, but false positives increased to 100 and specificity decreased to 0.6283. Threshold selection therefore created a sensitivity-focused operating point rather than improving every metric.

Error analysis identified both borderline and confident misclassifications. Grad-CAM indicated that the model frequently attended to breast regions, but activation near image borders and acquisition markers suggested possible reliance on non-clinical features. The visualisations could not confirm lesion localisation without quantitative comparison against annotations.

6.3 Limitations and Future Work

As discussed in Section 5.9, the findings should be interpreted in view of several limitations. Resizing the mammograms to 224 × 224 pixels may have removed clinically relevant details, while the use of one patient-level split and one completed training run for each configuration prevented the statistical reliability of small performance differences from being established. Furthermore, evaluation was conducted at image level, and external validation was not performed. The model and threshold-selection process should also be considered exploratory because ResNet50 was selected for further analysis after the comparative test evaluation.

Future work should therefore investigate higher-resolution or lesion-centred inputs, repeated runs or patient-level bootstrap confidence intervals, and evaluation at breast or patient level. Model and threshold selection should be completed using validation data before a single final test evaluation. Probability calibration, annotation-based Grad-CAM assessment, systematic error analysis by lesion and image characteristics, and external or cross-dataset validation would provide a more reliable assessment of model behaviour and generalisation.

6.4 Final Conclusion

This project demonstrates that architecture, fine-tuning, and classification threshold selection all influence mammography classification performance. No model was superior across every metric, and accuracy alone was insufficient for selecting the most appropriate configuration.

Within the scope of the experiments, fine-tuned ResNet50 provided the strongest overall performance at the default threshold. Applying the validation-selected threshold of 0.40 produced the strongest sensitivity-focused result, correctly identifying 79.40% of malignant images and reducing false negatives to 41. However, this improvement required accepting more false positives and lower specificity.

The research question can therefore be answered by concluding that the tested deep learning approaches distinguished benign and malignant mammograms with moderate effectiveness, although their suitability depended on the required balance between sensitivity and specificity. Fine-tuned ResNet50 at a threshold of 0.40 provided the most suitable sensitivity-focused configuration for prioritising malignant-case detection in this study.

Nevertheless, the remaining classification errors, limited uncertainty analysis, lack of external validation, and possible reliance on non-clinical image features mean that the

model is not suitable for clinical deployment. Instead, the project provides a technically grounded foundation for further research into sensitivity-focused, higher-resolution, calibrated, and externally validated mammography classification systems.

CHAPTER 7: REFERENCES

Abdelhafiz, D., Yang, C., Ammar, R., & Nabavi, S. (2019). Deep convolutional neural networks for mammography: Advances, challenges and applications. *BMC Bioinformatics*, *20*, Article 281. https://doi.org/10.1186/s12859-019-2823-4

Aggarwal, R., Sounderajah, V., Martin, G., Ting, D. S. W., Karthikesalingam, A., King, D., Ashrafian, H., & Darzi, A. (2021). Diagnostic accuracy of deep learning in medical imaging: A systematic review and meta-analysis. *npj Digital Medicine*, *4*(1), Article 65. https://doi.org/10.1038/s41746-021-00438-z

Ahmad, J., Akram, S., Jaffar, A., Ali, Z., Bhatti, S. M., Ahmad, A., & Rehman, S. U. (2024). Deep learning empowered breast cancer diagnosis: Advancements in detection and classification. *PLOS ONE*, *19*(7), Article e0304757. https://doi.org/10.1371/journal.pone.0304757

Carriero, A., Groenhoff, L., Vologina, E., Basile, P., & Albera, M. (2024). Deep learning in breast cancer imaging: State of the art and recent advancements in early 2024. *Diagnostics, 14*(8), 848. https://doi.org/10.3390/diagnostics14080848

Chollet, F. (2021). *Deep learning with Python* (2nd ed.). Manning Publications.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 770–778). https://doi.org/10.1109/CVPR.2016.90

Huang, G., Liu, Z., van der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 4700–4708). https://doi.org/10.1109/CVPR.2017.243

Jaamour, A., Myles, C., Patel, A., Chen, S. J., McMillan, L., & Harris-Birtill, D. (2023). A divide and conquer approach to maximise deep learning mammography classification accuracies. *PLOS ONE, 18*(5), e0280841. https://doi.org/10.1371/journal.pone.0280841

Keras Team. (n.d.). Keras documentation. Retrieved September 24, 2026, from https://keras.io/

Lee, R. S., Gimenez, F., Hoogi, A., Miyake, K. K., Gorovoy, M., & Rubin, D. L. (2016). *Curated Breast Imaging Subset of Digital Database for Screening Mammography (CBIS-DDSM)* [Data set]. The Cancer Imaging Archive. https://doi.org/10.7937/K9/TCIA.2016.7O02S9CY

Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., & Sánchez, C. I. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, *42*, 60–88. https://doi.org/10.1016/j.media.2017.07.005

Nasser, M., & Yusof, U. K. (2023). Deep learning based methods for breast cancer diagnosis: A systematic review and future direction. *Diagnostics*, *13*(1), Article 161. https://doi.org/10.3390/diagnostics13010161

Rodriguez-Ruiz, A., Lång, K., Gubern-Merida, A., Broeders, M., Hendriks, G., Samulski, M., de Baunt, V., Teuwen, J., Dahlblom, V., Wallis, M. G., Andersson, I., Zackrisson, S., & Sechopoulos, I. (2019). Stand-alone artificial intelligence for breast cancer detection in mammography. *Journal of the National Cancer Institute*, *111*(9), 916–922. https://academic.oup.com/jnci/article/111/9/916/5369731

Shen, D., Wu, G., & Suk, H.-I. (2017). Deep learning in medical image analysis. *Annual Review of Biomedical Engineering*, *19*, 221–248. https://doi.org/10.1146/annurev-bioeng-071516-044442

Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data*, *6*, Article 60. https://link.springer.com/article/10.1186/s40537-019-0197-0

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning* (Vol. 97, pp. 6105–6114). Proceedings of Machine Learning Research. https://proceedings.mlr.press/v97/tan19a.html

TensorFlow. (n.d.). TensorFlow documentation. Retrieved September 24, 2026, from https://www.tensorflow.org/

Wang, L. (2024). Mammography with deep learning for breast cancer detection. *Frontiers in Oncology*, *14*, Article 1281922. https://doi.org/10.3389/fonc.2024.1281922

World Health Organization. (2024, March 13). *Breast cancer*. https://www.who.int/news-room/fact-sheets/detail/breast-cancer

Appendix A: Source Code and Implementation Notebook

The complete source code and implementation notebook developed for this project are available at:

GitHub Repository:
https://github.com/cryst4lye0/CM3070-Deep-Learning-Breast-Cancer-Detection-Final-Project.git

The repository contains the final notebook used for data preprocessing, model development, training, evaluation, threshold analysis, misclassification analysis, and Grad-CAM visualisation.